

Design and implementation of motion tracking system based on IMU

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Abstract: The inertial measurement unit is widely used in the design and implementation of the motion tracking system. This paper designs a ICM20948-based motion tracking system that supports wireless transmission. Our system consists of 15 motion collection modules, data receiver and PC. The motion collection module is responsible for collecting, calibrating and fusing motion data. The data receiver packs the motion data and transmits it to the PC through the USB transmission module. To reduce the effect of system error on the motion data, we use the 6-sided calibration algorithm to calibrate the accelerometer, use ellipsoidal algorithm to calibrate the magnetometer, and use the deviation value to calibrate the gyroscope. An adaptive extended Kalman filter is used for fusion of motion data to obtain accurate attitude corner information. We use NRF24L01 to build a transmission hardware and innovatively propose a wireless data transmission mechanism to ensure the transmission integrity of data. Our motion tracking system supports wireless transmission and can ensure the real-time, synchronization and stability when collecting and transmitting motion data.

Key Words: Motion Tracking, Kalman Filter, Attitude Fusion

1 Introduction

Human motion tracking is a technique used to monitor and record human motion and convert the motion into attitude information. According to the different motion tracking position, it can be divided into the entire body's motion tracking, the tracking of the hand, and the tracking of the face [1–4]. This technology usually relies on the inertial sensor, camera and computer vision [5–9]. Motion tracking technology has a wide range of applications in different fields. Typical applications include that Daponte et al. [10] proposed a motion tracking system for family rehabilitation; Simonetto et al. [11] proposes a motion tracking system for virtual reality and augmented reality to build interaction with virtual reality; Acabado et al. [12] proposes an incentive treatment method based on the movement tracking system, which is used for the rehabilitation of the elderly.

At present, the advanced motion tracking system is designed based on the inertial measurement unit (IMU). The representative devices include the Moven inertial motion tracking system [13] from Netherlands and the 3DSuit inertial motion tracking system [14] from USA. The IMU-based motion tracking system has the advantages of low volume and low cost. In view of above advantages, we study the IMU-based motion tracking system. Firstly, the IMU's gyroscope and magne-

tometer will be affected by long-term accumulated errors and magnetic field environment respectively [15]. Therefore we use ellipsoidal algorithm to calibrate the magnetometer and use the deviation value to calibrate the gyroscope. Secondly, in order to fuse the motion data of the IMU multi-sensor, we use adaptive extended Kalman filter algorithm to achieve attitude fusion. Thirdly, in order to ensure the real-time and synchronization of the data under the condition of wireless transmission, we innovatively proposed the wireless data transmission mechanism. Our motion tracking system supports wireless transmission and can ensure the synchronization, real-time and integrity of motion data.

2 The Design of Hardware Circuit System

2.1 Motion Collection Module

Our motion tracking system tracks the key parts of the human body through 15 same motion collection modules (MCM). As shown in Fig. 1, each module corresponds to a key part, including head, back, arms, waist, thighs, calves and feet. The MCM can track the movement of each key part of the human body and accurately collect motion data. At the same time, the wireless transmission in the MCM can transmit motion data to the data receiver in real time. The MCM is composed of microcontroller, IMU, power supply module, power conversion circuit and wireless transmitter. We choose STM32G431CBU6 with strong versatility as the microcontroller. The IMU uses ICM20948 to balance the function and cost of the MCM. Lithium rechargeable battery and power conversion circuit are used for module power supply to ensure more lasting battery life at a limited volume. We use low-power NRF24L01 as the wireless transmitter, which can support long-term and stable transmission.

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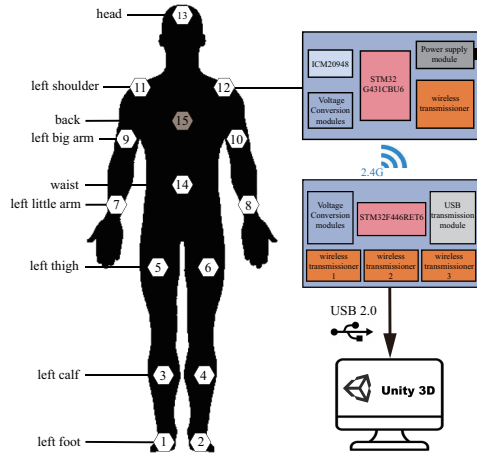


Fig. 1: The design of the motion tracking system. The motion tracking system includes MCM, data receiver and PC. 15 MCMs are worn on 15 parts of the human body, including head, shoulders, back, arms, waist, thighs, calfs and feet. Each MCM communicates with the data receiver through wireless transmission. Data receiver communicates to PC through USB transmission module.

2.2 Data Receiver

An important part of the motion tracking system is the data receiver, which is the data transmission bridge between PC and MCMs. The data receiver can receive the motion datas from 15 MCMs at the same time and pack them together. The data receiver is composed of microcontroller, wireless transmissioner and USB transmission module. We use a highly integrated and energy-saving chip STM32F446RET6 as the microcontroller. The wireless transmissioner is composed of 3 NRF24L01. This is because 2 NRF24L01 i and j can realize wireless communication when the sending register address of i is corresponding to the receiving register address of j . When NRF24L01 is used as a receiver, it can configure 6 different receiving register addresses. This enables it to communicate with 6 MCMs at the same time. Therefore we use 3 NRF24L01 ensure that the data receiver can receive the motion data from 15 MCMs at the same time. The USB transmission module uses USB2.0. Our hardware design of motion tracking equipment based on the wireless transmission can greatly improve the wearability of the equipment. The entire motion tracking system is shown in Fig. (1), MCM can collect the motion data of 15 key parts of the human body in real time. The wireless transmissioner are used to summarize 15 motion datas to the data receiver. Then the data is packed by the data receiver and transmitted to the PC through the USB transmission module.

3 Data Transmission Mechanism

The motion tracking system has high requirements for the synchronization and integrity of data. We prevent the loss of data packets by distinguishing channels. Specifically, we divid the 3 NRF24L01 in the data receiver into 3 channels, and divid 15 MCMS into these

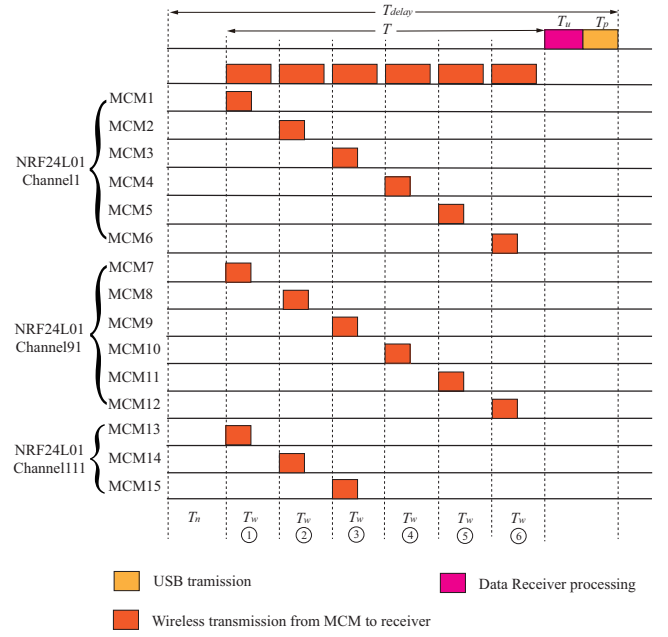


Fig. 2: Data transmission mechanism. The data transmission mechanism includes channel1, channel91 and channel111, where channel1 includes MCM1-6, channel91 includes MCM7-12 and channel111 includes MCM13-15. The delay of the system is T_{delay} , which consists of the duration of networking phase T_n , the cycle T of data frame, and the packaging processing time T_u and USB transmission time T_p . The cycle T is composed of 6 time gaps T_w .

3 channels. The 3 channels correspond to 6, 6 and 3 MCMs, respectively. As shown in Fig. (2), MCM1-6, MCM7-12 and MCM13-15 are divided into the channel1, channel91 and channel111, respectively. Furthermore, in order to ensure the real-time and synchronization of transmission, we use the real-time operating system FREERTOS. The FREERTOS allows the 3 NRF24L01 in the data receiver to request the motion data from the corresponding MCMs at the same time. The data receiver can integrate the motion data from 15 MCMs into a packet. A frame of data in the packet includes motion data, MCM address and channel lable. The transmission mechanism of motion tracking system is shown in Fig. 2. This mechanism not only effectively manages the sending and receiving time of motion data, but also ensures the correct sequence of motion data. Therefore, we ensure the reliability, synchronization and integrity of wireless transmission of system motion data by establishing a time transmission mechanism.

4 Data Calibration Based on IMU

The IMU of our motion tracking system is ICM20948. It is a 9-axis sensor, including 3-axis accelerometer, 3-axis gyroscope and 3-axis magnetometer. The accelerometer, gyroscope and magnetometer are used to measure the acceleration, angle and magnetic field direction of the 3-axis, respectively. ICM20948 has the advantages of high accuracy, low power and small size. The

MCM uses an analog IIC method to read the original data of ICM20948. The measurement accuracy of most IMUs in the market can guarantee within a high range. However, their measurement accuracy will be affected by manufacturing defects and environmental changes. Therefore, in order to achieve high measurement accuracy, we calibrate acceleration, angle and magnetic field direction of IMU, respectively.

4.1 Calibration of Accelerometer

The programmable range of acceleration meters plays a key role in measurement of acceleration. Large programmable range can effectively prevent measurement saturation, while small programmable range can improve the accuracy of measurement. Therefore, we choose accelerometer with a programmable range ($-8G, +8G$) to balance the range and accuracy of the measurement. Then we use 6-sided calibration algorithm to calibrate the accelerometer. The error model of the accelerometer is

$$\begin{bmatrix} a_{xreal} \\ a_{yreal} \\ a_{zreal} \end{bmatrix} = \begin{bmatrix} S_x & C_{xy} & C_{xz} \\ C_{yx} & S_y & C_{yz} \\ C_{zx} & C_{zy} & S_z \end{bmatrix} \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}, \quad (1)$$

where a is the acceleration on each axis, S is the proportional factor error and C is the cross error. To eliminate the error of the accelerometer, we increase the deviation b on each axis. Specifically, we collect the positive and negative direction data of the 3 axis of the accelerometer respectively. The deviation is $b = 1/2(b^+ + b^-)$, where b^+ and b^- is the extreme value of the positive and negative direction data of axis respectively. Then we have

$$\begin{bmatrix} a_{xreal} \\ a_{yreal} \\ a_{zreal} \end{bmatrix} = \begin{bmatrix} S_x & C_{xy} & C_{xz} \\ C_{yx} & S_y & C_{yz} \\ C_{zx} & C_{zy} & S_z \end{bmatrix} \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} + \begin{bmatrix} b_{ax} \\ b_{ay} \\ b_{az} \end{bmatrix}. \quad (2)$$

The error model after deviation calibration can obtain relatively accurate data. The processed data is then converted to accelerated data. In order to obtain smooth results, we use the method of threshold processing for accelerated data, where the threshold is 0.1. Specifically, when the difference between the current acceleration data and the previous data is less than the threshold, we use the previous data to replace the current data. This can avoid unnecessary data fluctuations caused by sensor noise or small changes. The comparison of calibrated acceleration data and original data is shown in Fig. 3(a), it's obvious that the calibrated data is more accurate and smooth.

4.2 Calibration of Magnetometer

Most of the magnetometer calibration of IMU uses ellipsoid fitting methods [16]. The ellipsoid fitting algorithm can convert magnetometer data into real magnetic fields. This algorithm usually fits the measurement results of the magnetometer with an ellipsoid model. The optimal result of the fitting parameters is then found by the minimum two multiplication. The

standard equation of the ellipsoid is

$$\left(\frac{X - X0}{A}\right)^2 + \left(\frac{Y - Y0}{B}\right)^2 + \left(\frac{Z - Z0}{C}\right)^2 = 1, \quad (3)$$

where $X0$, $Y0$ and $Z0$ represent the central coordinate value of the oval ball. A , B and C represent the half axis length. The ellipsoid fitting process of magnetometer includes data collection, data conversion, and ellipsoid fit. Firstly, we get 2500 original data through MCM. Secondly, we convert the original data into magnetic data and perform a meanization processing. Thirdly, through the ellipsoid fitting, we get the center coordinates of the ellipsoid $X0 = 1.767611344$, $Y0 = 0.785945815$ and $Z0 = 1.637230958$, the shaft radius $A = 54.094$, $B = 58.961$ and $C = 58.924$ and the scaling factor $KA = 1.0597$, $KB = 0.9722$ and $KC = 0.9728$. We bring the fitting parameters into Equ. (3) to obtain the fitting results, which are shown in Fig 3(b). It can be seen that the ellipsoid after fitting is near the coordinate center point. In order to reduce the impact of environmental changes on magnetic data, we conduct threshold processing on the fitted magnetometer data, where the threshold is 0.1. The comparison of calibrated magnetometer data and original data is shown in Fig. 3(c). The results clearly show that the data after calibration is smoother and stable.

4.3 Calibration of Gyroscope

The calibration process of the gyroscope is similar to that of the calibration process of the accelerometer. We need to calculate the deviation value to calibrate the error of the gyroscope. The comparison of calibrated gyroscope data and original data is shown in Fig. 3(d), it is found that the data of the 3 axes of the gyroscope after calibration are maintain at 0. This shows that the result of the calibration is more stable.

5 AEKF for Attitude Fusion

Attitude fusion refers to the fusion of different sensors data to obtain more accurate and more stable attitude information. Each sensor has its advantages and limitations. Through the fusion of the attitude, the advantages of each sensor can be comprehensively utilized while reducing their individual shortcomings. Common attitude fusion algorithms include Kalman filter [17], complementary filter [18], Madgwick AHRS filter [19] and particle filtering [20].

The Kalman filter algorithm is widely used in attitude fusion [21]. It can integrate the data of multiple sensors and get the optimal estimate of the attitude fusion result. The extended Kalman filter algorithm was proposed to adapt to a non-linear system [22]. Adaptive extended Kalman filter (AEKF) algorithm [23] can adjust the parameters according to different situations, aiming to adapt to various noise environments. Therefore, we use the AEKF algorithm for attitude fusion. We use quaternion $Q = [q_0, q_1, q_2, q_3]^T$ in the AEKF algorithm to represent the space direction. The AEKF algorithm is divided into two processes: forecast and correction. The gyroscope data is used to predict, and

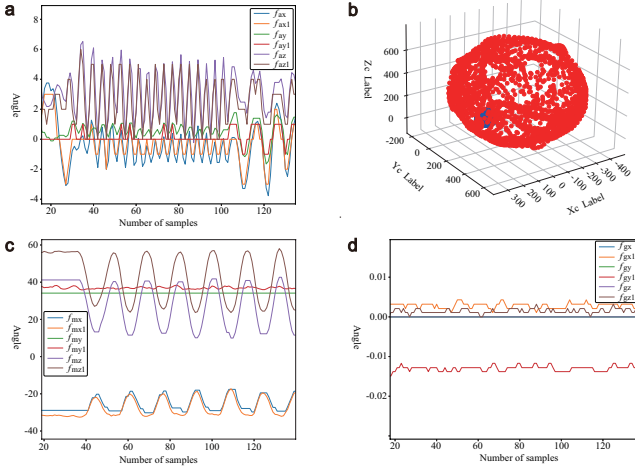


Fig. 3: Data Calibration Based on IMU. Comparison of original data (f_{ax1} , f_{ay1} , f_{az1} , f_{mx1} , f_{my1} , f_{mz1} , f_{gx1} , f_{gy1} and f_{gz1}) and calibrated data (f_{ax} , f_{ay} , f_{az} , f_{mx} , f_{my} , f_{mz} , f_{gx} , f_{gy} and f_{gz}) for the (a) accelerometer, the (c) gyroscope and the (d) magnetometer. (b) The fitting results of the original data based on the ellipsoid fitting method, where the red points are the original data and the blue points are the fitting results.

the data of accelerometer and magnetometer are used for correction. The steps of the AEKF algorithm is shown in Table 1.

Table 1: Adaptive extended Kalman filter process. The state transfer matrix is A , the process noise collaboration matrix is S , the observation transfer matrix is A , the Kalman Gain is K , the observation matrix is G and the error collaboration matrix is P . Q represents a state matrix at a certain moment, which is converted through quaternion $Q = [q_0, q_1, q_2, q_3]^T$. I represents the unit matrix.

steps	formula
1	$\hat{Q}_{k/k-1} = A Q_{k-1}$
2	$\hat{P}_{k/k-1} = A P_{k-1} A^T + S$
3	$\hat{K}_k = A Q_{k-1} \frac{P_{k/k-1} H^T}{H P_{k/k-1} H^T + R}$
4	$\hat{Q}_k = Q_{k/k-1} + K_k (G_k - H Q_{k/k-1})$
5	$\hat{P}_k = (I - K_k H) P_{k/k-1}$

Specifically, after determining the system's status prediction equation (step1) and covariance prediction equation (step2), the system state prediction is implemented. We obtain the best estimate of the state matrix Q in real time by iterating steps1-5 in Table 1. Then we obtain the fused attitude angle results pitch, roll and yaw through the following equations.

$$pitch = \text{asin}(2(q_0 q_2 - q_3 q_1)), \quad (4)$$

$$roll = \text{atan2}(2(q_0 q_1 + q_2 q_3), 1 - 2(q_1^2 + q_2^2)), \quad (5)$$

$$yaw = \text{atan2}(2(q_0 q_3 + q_1 q_2), 1 - 2(q_2^2 + q_3^2)). \quad (6)$$

AEKF can adapt to the quaternion, adjust the error of the observation value and predictive values. Therefore, the result of a smooth attitude can be obtained.

Based on the 800 groups of randomly collected motion data of MCM, we compare the results of fusion of the AEKF algorithm and the Madgwick AHRS filtering algorithm. The comparison results are shown in Fig. 4, we found that the results obtained by AEKF algorithm are smoother. It is worth noting that the 800 groups of motion data we have collected are obtained under the conditions of MCM-based fast and significant movement. As shown in Fig. 4, compared to the Madgwick AHRS filtering algorithm, the AEKF algorithm can get relatively smooth results.

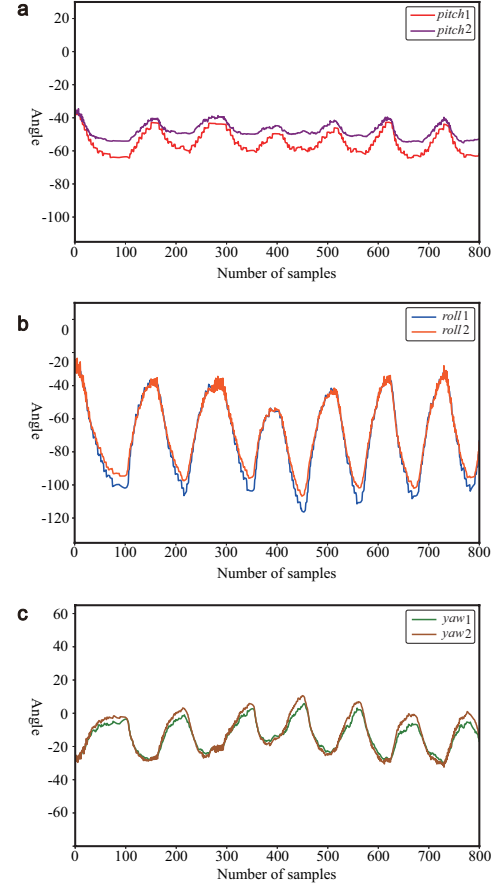


Fig. 4: The result of the fusion of the attitude. Comparison of AEKF's attitude fusion result ($pitch1$, $roll1$ and $yaw1$) and Madgwick AHRS filtering's attitude fusion result ($pitch2$, $roll2$ and $yaw2$) for the (a) pitch angle, the (b) roll angle and the (c) yaw angle.

6 Conclusion

In this paper, a motion tracking system based on ICM20948 is designed, which consists of 15 MCMs, data receiver and PC. The MCM is responsible for collecting, calibrating and fusing motion data. Each MCM transmits the collected motion data to the data receiver through wireless transmission, avoiding the motion angle restrictions caused by wired devices and greatly improving the wearability of the device. The data receiver packs the motion data and transmits it to the PC through the USB transmission module. In order to reduce the effect of IMU system error on the measurement result, we use the 6-sided calibration algorithm to

calibrate accelerometer, use ellipsoidal algorithm to calibrate the magnetometer, and use the deviation value to calibrate the gyroscope. We can obtain more stable and smooth results by comparing the motion data before and after calibration. The AEKF algorithm is used for attitude fusion and adaptive adjustment to make the data smoother. We use 3 NRF24L01 as the receiving module of the data receiver, and innovatively proposed the data transmission mechanism based on distinguishing channels to ensure the integrity of the motion data. Our motion tracking system supports wireless transmission, which can ensure the real-time and stability of the collection and transmission of motion data.

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